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Atomic oscillator, control method, and control apparatus

Abstract

An atomic oscillator of the present disclosure includes: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a controller that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies. Then, the two gas cells are configured to have different characteristics of the corresponding resonance frequencies with respect to temperature, and the controller is configured to control the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

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U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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2017-011680	12/2016	JP	N/A

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from Japanese patent application No. 2023-013282, filed on Jan. 31, 2023, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present invention relates to an atomic oscillator, a control method, a control apparatus, and a program.

BACKGROUND ART

(3) An atomic oscillator is an apparatus that measures accurate time based on the natural frequency of an atom. A compact atomic clock mainly measures the natural frequency of an atom using the CPT (Coherent Population Trapping), which is a quantum interference effect occurring when an alkali metal atomic gas is irradiated with excitation lights having two frequencies, as the oscillation principle of an atomic oscillator. In the CPT, when the difference between the frequencies of the two excitation lights matches the transition frequency between the ground levels of the alkali metal, the excitation light is not absorbed and the amount of transmitted light becomes larger. The atomic oscillator using the CPT as the operation principle sweeps the difference between the frequencies of the two excitation lights, and determines a resonance frequency, which is the difference between the frequencies that maximizes the amount of transmitted light, as the natural frequency of the

atom. It is one of the performance indexes of the atomic oscillator whether or not the natural frequency of the atom can be stably acquired for a long term.

(4) In the atomic oscillator described above, one factor of performance degradation is a temperature shift, which is fluctuation of the resonance frequency caused by a temperature change in the alkali metal atomic gas. That is to say, when a temperature change inside the oscillator occurs, the optical transition characteristic of the atom fluctuates, and the stability of the oscillation frequency decreases. In response to such a problem, Patent Literature 1 describes that a temperature measuring device and a heater are provided outside an alkali metal cell. Then, Patent Literature 1 describes that the temperature of the alkali metal cell is kept constant by heating the alkali metal cell with the heater using the temperature information of the cell. Patent Literature 1: Japanese Unexamined Patent Application Publication No. JP-A 2017-011680

(5) However, the atomic oscillator of Patent Literature 1 described above firstly causes a problem that the power consumption increases for heating the heater. Moreover, in a case where the ambient temperature of the alkali atom cell is measured by the temperature measuring device, the temperature of the atom itself is not measured, and the frequency stability cannot be increased more. Furthermore, in the atomic oscillator of Patent Literature 1, wiring for the heater and the temperature measuring device is required, and a magnetic field emitted by the wiring may affect the frequency, so that the frequency stability cannot be increased more.

SUMMARY OF THE INVENTION

(6) Accordingly, an object of the present disclosure is to provide an atomic oscillator that can solve the abovementioned problem that it is difficult to suppress the power consumption and further increase the frequency stability.

(7) An atomic oscillator as an aspect of the present disclosure includes: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a controller that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies. The two gas cells are configured to have different characteristics of the corresponding resonance frequencies with respect to temperature. The controller controls the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

(8) A control method as an aspect of the present disclosure is a control method by an atomic oscillator. The atomic oscillator includes: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a control apparatus that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies. The two gas cells are configured to have different characteristics of the corresponding resonance frequencies with respect to temperature. The control method includes, by the control apparatus, controlling the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

(9) A control apparatus as an aspect of the present disclosure is a control apparatus of an atomic oscillator. The atomic oscillator includes: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a control apparatus that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined

resonance frequencies. The two gas cells are configured to have different characteristics of the corresponding resonance frequencies with respect to temperature. The control apparatus controls the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

(10) A program as an aspect of the present disclosure is a computer program of an atomic oscillator. The atomic oscillator includes: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a control apparatus that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies. The two gas cells are configured to have different characteristics of the corresponding resonance frequencies with respect to temperature. The computer program includes instructions for causing the control apparatus to processes to control the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

(11) With the configuration as described above, the present disclosure can provide an atomic oscillator that can further increase the frequency stability while suppressing the power consumption.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram showing the configuration of an atomic oscillator in a first example embodiment of the present disclosure;

(2) FIG. 2 is a view showing signals output from the respective components of the atomic oscillator disclosed in FIG. 1;

(3) FIG. 3 is a view showing the characteristic of the atomic oscillator disclosed in FIG. 1;

(4) FIG. 4 is a view showing the characteristic of the atomic oscillator disclosed in FIG. 1;

(5) FIG. 5 is a view showing processing by the atomic oscillator disclosed in FIG. 1;

(6) FIG. 6 is a flowchart showing the processing operation of the atomic oscillator disclosed in FIG. 1;

(7) FIG. 7 is a block diagram showing the configuration of an atomic oscillator in a second example embodiment of the present disclosure; and

(8) FIG. 8 is a flowchart showing the operation of the atomic oscillator in the second example embodiment of the present disclosure.

EXAMPLE EMBODIMENT

First Example Embodiment

(9) A first example embodiment of the present disclosure will be described with reference to FIGS. 1 to 6. FIGS. 1 to 5 are views for describing the configuration of an atomic oscillator, and FIG. 6 is a view for describing the operation of the atomic oscillator.

(10) [Configuration]

(11) FIG. 1 is a block diagram showing the configuration of an atomic oscillator in this example embodiment. The atomic oscillator in this example embodiment includes a light generator 1, two gas cells 21 and 22, a light detector 3, a controller 4, a VCXO (Voltage Controlled Crystal Oscillator) 51, a frequency converter 52, and a current modulator 53. The light generator 1 includes a laser 11. The controller 4 includes a resonance frequency and temperature shift amount calculator 41, a current modulator controller 42, a laser controller 43, a frequency converter controller 44, and a VCXO controller 45. The controller 4 is configured by an information processing apparatus

(control apparatus) including an arithmetic logic unit and a memory unit, and the respective parts **41** to **45** of the controller **4** are realized by execution of a program by the arithmetic logic unit. The respective components will be described in detail below.

(12) The light generator **1**, in cooperation with the controller **4**, the VCXO **51**, the frequency converter **52** and the current modulator **53**, generates two excitation lights (irradiation lights) including at least two different frequency components by performing frequency modulation from a single wavelength excitation light, and irradiates the two gas cells **21** and **22**, respectively. FIG. 2 shows signals output from the respective components of the atomic oscillator. As shown in this figure, a current modulation frequency conversion coefficient and an output frequency conversion coefficient that are set in advance are output from the frequency converter controller **44** and, by output of a VCXO oscillation voltage based on the output frequency conversion coefficient from the VCXO controller **45**, the oscillation frequency of an output frequency signal that is output from the VCXO **51** and becomes an external output is controlled. Moreover, the output frequency signal output from the VCXO **51** is subject to frequency conversion by the frequency converter **52** based on the current modulation frequency conversion coefficient and, for example, a current modulation frequency signal around 9.2 GHz corresponding to the transition frequency between the ground levels of a cesium atom is output. Then, a laser drive current output from the laser controller **43** is modulated by the current modulator **53** based on a setting value of the current modulator **53** output from the current modulator controller **42**, and the laser **11** is driven by the modulated laser drive current. Consequently, two excitation lights having different frequencies are generated by the light generator **1**, the difference between the frequencies of the two excitation lights is swept around 9.2 GHz by changing the setting value of the current modulator **53**, and the excitation lights are applied to the two gas cells **21** and **22**, respectively. Since the configuration and method for generating an excitation light by the light generator **1** of the atomic oscillator are known, a detailed description thereof will be omitted.

(13) Herein, the values of “current modulation frequency conversion coefficient” and “output frequency conversion coefficient” described above are set in advance, respectively, and are stored in the frequency converter controller **44**. However, in this example embodiment, the temperature shift amount of a resonance frequency is calculated as will be described later, and the values of “current modulation frequency conversion coefficient” and “output frequency conversion coefficient” are changed to values corresponding to the temperature shift amount. This will be discussed later.

(14) Next, the two gas cells **21** and **22** will be described. The atomic oscillator in this example embodiment includes two gas cells with alkali metal atoms encapsulated in. Specifically, first, the atomic oscillator includes a frequency measuring gas cell **21**, which is a gas cell used for measuring the natural frequency of an atom using the CPT as the oscillation principle of the atomic oscillator as described before. In addition to this, the atomic oscillator includes a temperature measuring gas cell **22** in this example embodiment.

(15) Herein, the alkali metal atoms encapsulated in the two gas cells **21** and **22** may be any alkali metal atoms such as cesium atoms, rubidium atoms, sodium atoms, and potassium atoms. Meanwhile, the two gas cells **21** and **22** are configured so that characteristics of corresponding resonance frequencies with respect to temperature are different from each other. As an example, FIG. 3 shows the relation between the shift amount (fluctuation amount) of resonance frequency and temperature in a case where the alkali metal atoms are cesium atoms and the buffer gas is nitrogen, and an equation for calculating the shift amount $f_{\text{sub.Res}}$ (Hz) of resonance frequency is shown in Equation 1.

(16) $f_{\text{Res}} = P(T + T^2) + c$ [Equation1] P: gas pressure (torr) T: temperature (° C.) δ : 0.824 Hz/(° C..Math.torr) for nitrogen γ : -0.00251 Hz/(° C..sup.2.Math.torr) for nitrogen c: constant

(17) The above δ , γ and c are values determined by the type of the buffer gas. Therefore, by

differentiating the types and gas pressures of buffer gases encapsulated in the frequency measuring gas cell **21** and the temperature measuring gas cell **22**, the two gas cells **21** and **22** can be configured to have different fluctuation amounts of resonance frequencies with respect to temperature change. In this example embodiment, as shown in FIG. 4, the frequency measuring gas cell **21** and the temperature measuring gas cell **22** are configured so that the types of the buffer gases encapsulated therein are different from each other. As an example, a 19 torr, Ar/N.sub.2 mixed gas (P.sub.Ar/P.sub.N2=0.62) shown by reference character A1 in FIG. 4 is used for the frequency measuring gas cell **21**, and a 30 torr, N.sub.2 gas shown by reference character A2 in FIG. 4 is used for the temperature measuring gas cell **22**. The buffer gases shown by reference characters A1 and A2 in FIG. 4 are described in the paper "Temperature and pressure shift of the Cs clock transition in the presence of buffer gases: Ne, N2, Ar, Phys. Rev. A83, 062714". Moreover, cesium is encapsulated in both the gas cells. Consequently, the fluctuation amount of resonance frequency with respect to temperature change at 25 to 50° C. in the frequency measuring gas cell **21** becomes "−0.6 Hz/° C.", the fluctuation amount of resonance frequency with respect to temperature change at 25 to 50° C. in the temperature measuring gas cell **22** becomes "20 Hz/° C.", and the temperature dependences of the respective resonance frequencies are differentiated.

(18) The light detector **3** detects transmitted lights transmitted by the two gas cells **21** and **22** described above, and measures the amounts of the transmitted lights. At this time, since the excitation lights are applied while the setting values in frequency modulation are being changed, the light detector **3** passes the measured transmitted light amounts to the controller **4** so that the measured transmitted light amounts are recorded in association with the changed setting values, that is, the swept frequency differences. Thus, as indicated by reference character B1 in FIG. 5, the light detector **3** measures so that a graph of transmitted light amount with respect to frequency difference corresponding to the frequency measuring gas cell **21** (graph denoted by reference character f1 on the left side) and a graph of transmitted light amount with respect to frequency difference corresponding to the temperature measuring gas cell **22** (graph denoted by reference character f2 on the right side) are recorded by the controller **4**.

(19) The resonance frequency and temperature shift amount calculator **41** of the controller **4** determines resonance frequencies corresponding to the respective gas cells **21** and **22** based on the transmitted light amounts at the respective gas cells **21** and **22** passed from the light detector **3**. That is to say, the resonance frequency and temperature shift amount calculator **41** determines a resonance frequency f1 corresponding to the frequency measuring gas cell **21** and a resonance frequency f2 corresponding to the temperature measuring gas cell **22**. For example, the resonance frequency and temperature shift amount calculator **41** may determine difference frequencies at which the transmitted light amounts become maximum, respectively, in the respective graphs indicated reference character B1 in FIG. 5 as the resonance frequencies f1 and f2, or may determine difference frequencies at which the slopes of the transmitted light amount graphs with respect to frequency difference become zero by differentiated waveform measurement of the transmitted light amounts as the resonance frequencies.

(20) Then, the controller **4** controls the oscillation frequency based on the determined resonance frequency f1 corresponding to the frequency measuring gas cell **21**. Specifically, the controller **4** controls, in the VCXO controller **45**, the oscillation frequency of an output frequency signal output by the VCXO **51** based on the resonance frequency f1 corresponding to the frequency measuring gas cell **21** (one gas cell) and an output frequency conversion coefficient (first parameter) stored in advance in the frequency converter controller **44**. As an example, let us assume that an output frequency signal of 10 MHz is output by the VCXO **51** in the atomic oscillator, and that the determined resonance frequency f1 of the frequency measuring gas cell **21** is "9.192631 GHz" and a preset output frequency conversion coefficient is "(10 MHz)/(9.192631 GHz)". In this case, an output frequency signal of (9.192631 GHz)*(10 MHz)/(9.192631 GHz)=10 MHz is output by the VCXO **51**. Consequently, the output frequency signal of 10 MHz is output externally from the

atomic oscillator, and is also input into the frequency converter **52**. Moreover, at this time, assuming that a preset current modulation frequency conversion coefficient (second parameter) is “ $(9.192631 \text{ GHz})/(10 \text{ MHz})$ ”, a current modulation frequency signal of $(10 \text{ MHz}) \times (9.192631 \text{ GHz})/(10 \text{ MHz}) = 9.192631 \text{ GHz}$ is generated by the frequency converter **52**, that is, a reference signal to be the reference for the irradiation light is generated, and is input into the current modulator **53**. The values of the output frequency conversion coefficient and the current modulation frequency conversion coefficient are preset and stored based on the oscillation frequency of the output frequency signal assumed to be output by the VCXO **51** in the atomic oscillator and the characteristics of resonance frequency shift amount and temperature of the frequency measuring gas cell **21** and the temperature measuring gas cell **22** as shown in FIG. **4**.

(21) The controller **4** in this example embodiment is further configured to control the oscillation frequency based on the results of comparison between the determined resonance frequencies f_1 and f_2 of the frequency measuring gas cell **21** and the temperature measuring gas cell **22** as will be described below, in order to cope with fluctuations in the resonance frequencies f_1 and f_2 due to temperature changes in the frequency measuring gas cell **21** and the temperature measuring gas cell **22**.

(22) The resonance frequency and temperature shift calculator **41** calculates the difference between the determined resonance frequencies f_1 and f_2 of the frequency measuring gas cell **21** and the temperature measuring gas cell **22**, and calculates a temperature shift amount, which is the fluctuation amount of resonance frequency with respect to temperature change, based on the difference between the resonance frequencies f_1 and f_2 . Specifically, first, the resonance frequency and temperature shift amount calculator **41** stores in advance a graph representing the relation between difference $(f_2 - f_1)$ in shift amounts of the resonance frequencies f_1 and f_2 of the frequency measuring gas cell **21** and the temperature measuring gas cell **22** and temperature T as indicated by reference character B2 in FIG. **5**. The graph as indicated by reference character B2 in FIG. **5** is created and stored in advance based on the characteristics of resonance frequency shift amount and temperature of the frequency measuring gas cell **21** and the temperature measuring gas cell **22** as indicated by reference characters A1 and A2 in FIG. **4**. In addition to this, the resonance frequency and temperature shift amount calculator **41** stores a graph representing the relation between temperature T and the temperature shift amount that is the fluctuation amount of each of the resonance frequencies f_1 and f_2 of the frequency measuring gas cell **21** and the temperature measuring gas cell **22**. Specifically, the graph of the temperature shift amount to temperature T is represented by a correspondence table of temperature T and output frequency conversion coefficient and a correspondence table of temperature T and current modulation frequency conversion coefficient, and the correspondence tables are generated and stored in advance. The correspondence table between temperature T and output frequency conversion coefficient and the correspondence table between temperature T and current modulation frequency conversion coefficient are generated and stored in advance based on the oscillation frequency of an output frequency signal that is assumed to be output by the VCXO **51** in the atomic oscillator and the characteristics of resonance frequency shift amounts and temperature of the frequency measuring gas cell **21** and the temperature measuring gas cell **22** as shown in FIG. **4**.

(23) Consequently, the resonance frequency and temperature shift amount calculator **41** identifies temperature T from the difference $(f_1 - f_2)$ in the resonance frequencies f_1 and f_2 of the frequency measuring gas cell **21** and the temperature measuring gas cell **22** having been measured, based on the stored graphs and correspondence tables, and specifies an output frequency conversion coefficient and a current modulation frequency conversion coefficient from the temperature T . The resonance frequency and temperature shift amount calculator **41** performs temperature correction by changing and updating the output frequency conversion coefficient and the current modulation frequency conversion coefficient stored in the frequency converter controller **44** to the newly specified output frequency conversion coefficient and current modulation frequency conversion

coefficient.

(24) Then, the controller **4** controls the oscillation frequency based on the determined resonance frequency **f1** corresponding to the frequency measuring gas cell **21** in the same manner as described above by using the output frequency conversion coefficient and current modulation frequency conversion coefficient updated by the temperature correction. Herein, as an example, assuming that the resonance frequency **f1** of the frequency measuring gas cell **21** fluctuates from “9.192631 GHz” to “9.192632 GHz” due to a temperature shift, a case where temperature correction on the output frequency conversion coefficient and the current modulation frequency conversion coefficient described above is not performed, and a case where the temperature correction is performed will be described.

(25) First, in a case where the temperature correction is not performed, the output frequency conversion coefficient remains “ $(10 \text{ MHz})/(9.192631 \text{ GHz})$ ”, and an output frequency signal of $(9.192632 \text{ GHz}) \times (10 \text{ MHz}) / (9.192631 \text{ GHz}) = 10.000011 \text{ MHz}$ will be output by the VCXO **51** using the temperature-shifted resonance frequency (9.192632 GHz) of the frequency measuring gas cell **21**. The current modulation frequency conversion coefficient remains “ $(9.192631 \text{ GHz})/(10 \text{ MHz})$ ”, and a current modulation frequency signal of $(10.000011 \text{ MHz}) \times (9.192631 \text{ GHz}) / (10 \text{ MHz}) = 9.192632 \text{ GHz}$ is output by the frequency converter **52**. For this reason, the atomic oscillator should have performed frequency modulation of the excitation light at “9.192631 GHz” and externally output an output frequency signal of “10 MHz”, but in fact, the atomic oscillator performs frequency modulation at “9.192632 GHz” and externally outputs an output frequency signal of “10.000011 MHz”.

(26) On the other hand, in a case where the temperature correction is performed, the output frequency conversion coefficient is updated to “ $(10 \text{ MHz})/(9.192632 \text{ GHz})$ ”, and an output frequency signal of $(9.192632 \text{ GHz}) \times (10 \text{ MHz}) / (9.192632 \text{ GHz}) = 10 \text{ MHz}$ is output by the VCXO **51** using the temperature-shifted resonance frequency (9.192632 GHz) of the frequency measuring gas cell **21**. The current modulation frequency conversion coefficient is updated to “ $(9.192632 \text{ GHz})/(10 \text{ MHz})$ ”, and a current modulation frequency signal of $(10 \text{ MHz}) \times (9.192632 \text{ GHz}) / (10 \text{ MHz}) = 9.192632 \text{ GHz}$ is output by the frequency converter **52**. For this reason, the atomic oscillator performs frequency modulation of the excitation light at the resonance frequency of “9.192632 GHz” and externally outputs an output frequency signal of “10 MHz”.

(27) Instead of the correspondence table of temperature **T** and output frequency conversion coefficient and the correspondence table of temperature **T** and current modulation frequency conversion coefficient described above, the controller **4** may store a correspondence table of the difference between the resonance frequencies **f1** and **f2** and output frequency conversion coefficient or current modulation frequency conversion coefficient, and may store a correspondence table of the difference between the resonance frequencies **f1** and **f2** and fluctuation amounts of the respective resonance frequencies **f1** and **f2**, and a formula for calculating the fluctuation amounts of the respective resonance frequencies **f1** and **f2** from the difference between the resonance frequencies **f1** and **f2**. Then, the controller **4** may calculate the temperature shift amounts of the respective resonance frequencies **f1** and **f2** described above from the difference between the resonance frequencies based on these correspondence tables and calculation formula, and control the oscillation frequency based on the values.

(28) [Operation]

(29) Next, the operation of the atomic oscillator described above, particularly, the operation when correcting a temperature shift in resonance frequency will be described with reference to a flowchart of FIG. 6.

(30) First, the controller **4** sets various setting values, and the light generator **1** generates an excitation light having at least two different frequency components. At this time, the current modulator controller **42** of the controller **4** sweeps a modulation frequency while changing around 9.2 GHz that corresponds to the transition frequency between the ground levels of a cesium atom,

for example (step S1).

(31) Then, the light generator **1** applies the generated excitation light to the two gas cells **21** and **22**, respectively, and the light detector **3** detects transmitted lights (step S2). At this time, the controller **4** stores the measured amounts of the transmitted lights in association with the modulation frequency (step S3). Then, the atomic oscillator repeats the above process until the sweep of the modulation frequency and the measurement of the transmitted lights are completed (No at step S4, Step S7).

(32) The controller **4** performs temperature correction in the following manner at any timing after the measurement of the transmitted lights at the swept modulation frequency is completed (Yes at step S4). First, the controller **4** determines the resonance frequencies f_1 and f_2 corresponding to the respective gas cells **21** and **22**, based on the transmitted light amounts corresponding to the respective gas cells **21** and **22** passed from the light detector **3**. Then, the controller **4** calculates the difference between the determined resonance frequencies f_1 and f_2 of the frequency measuring gas cell **21** and the temperature measuring gas cell **22**, and calculates a temperature shift amount, which is the amount of fluctuation of resonance frequency to temperature change, based on the difference between the resonance frequencies f_1 and f_2 (step S5). For example, the controller **4** identifies temperature T based on a graph showing the relation between the difference in the shift amounts of the resonance frequency f_1 of the frequency measuring gas cell **21** and the resonance frequency f_2 of the temperature measuring gas cell **22** and temperature T as indicated by reference character **B2** in FIG. 5. Furthermore, the controller **4** specifies an output frequency conversion coefficient and a current modulation frequency conversion coefficient corresponding to the identified temperature T , based on a correspondence table of temperature T and output frequency conversion coefficient and a correspondence table of temperature T and current modulation frequency conversion coefficient.

(33) Then, the controller **4** changes and updates the stored output frequency conversion coefficient and current modulation frequency conversion coefficient to the newly specified output frequency conversion coefficient and current modulation frequency conversion coefficient (step S6). After that, by controlling an oscillation frequency from the VCXO **51** using the output frequency conversion coefficient and the current modulation frequency conversion coefficient updated by the temperature correction, the atomic oscillator can appropriately cope with the temperature shift.

(34) Thus, with the atomic oscillator of the present disclosure, it is possible to appropriately cope with a temperature shift in which a resonance frequency fluctuates due to a temperature change of an alkali metal atomic gas and increase frequency stability. Unlike in Patent Literature 1, the ambient temperature of a gas cell is not measured by an additional device and there is no need to use a heater, so that it is possible to further increase frequency stability while inhibiting increase of wiring and power consumption. That is to say, according to the present invention, by setting a parameter on the oscillation frequency based on the difference between the resonance frequencies corresponding to the two gas cells, it is possible when a temperature shift in which the resonance frequencies fluctuate due to a temperature change to control the oscillation frequency in response to the shift amount, and it is possible to increase frequency stability.

Second Example Embodiment

(35) Next, a second example embodiment of the present invention will be described with reference to FIGS. 7 and 8. FIG. 7 is a block diagram showing the configuration of an atomic oscillator in the second example embodiment, and FIG. 8 is a flowchart showing the operation of the atomic oscillator. In this example embodiment, the overview of the configurations of the atomic oscillator and the control method described in the example embodiment described above.

(36) As shown in FIG. 7, an atomic oscillator **100** in this example embodiment includes: two gas cells **101a** and **101b** in which alkali metal atoms are encapsulated, respectively; a light generator **102** that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector **103** that detects transmitted lights transmitted by the two gas cells, respectively; and a controller **104** that determines resonance frequencies corresponding to

the respective two gas cells based on the light amounts of the transmitted lights by the respective two gas cells and controls an oscillation frequency based on the determined resonance frequencies. Then, the two gas cells **101a** and **101b** are configured so that characteristics of the corresponding resonance frequencies with respect to temperature are different from each other, and moreover, the controller **104** is configured to control the oscillation frequency based on the result of comparison between the resonance frequencies corresponding to the respective two gas cells.

(37) Then, the controller **104** (control device) with the above configuration executes processes to: determine resonance frequencies in the two gas cells **101a** and **101b** having different characteristics of resonance frequency with respect to temperature, respectively (step **S101**); and control the oscillation frequency based on the result of comparison between the resonance frequencies corresponding to the respective two gas cells (step **S102**) as shown in FIG. **8**. The processes by the controller **104** are realized by execution of a program by an arithmetic logic unit configuring the controller **104**.

(38) Thus, in the atomic oscillator of the present disclosure, the oscillation frequency is controlled based on the result of comparison between the resonance frequencies of two gas cells having different characteristics of resonance frequency with respect to temperature. Consequently, it is possible to further increase frequency stability while inhibiting increase of components and power consumption in the atomic oscillator.

(39) Although the present invention has been described above with reference to the example embodiments and the like disclosed above, the present invention is not limited to the example embodiments described above. The configurations and details of the present invention can be changed in various manners that can be understood by one skilled in the art within the scope of the present invention.

SUPPLEMENTARY NOTES

(40) The whole or part of the example embodiments disclosed above can be described as the following supplementary notes. Below, the overview of the configurations of the atomic oscillator, the control method, the control apparatus, and the program according to the present invention will be described. Meanwhile, the present invention is not limited to the following configurations.

Supplementary Note 1

(41) An atomic oscillator comprising: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a controller that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies, wherein: the two gas cells are configured to have different characteristics of the corresponding resonance frequencies with respect to temperature; and the controller controls the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

Supplementary Note 2

(42) The atomic oscillator according to Supplementary Note 1, wherein the controller controls the oscillation frequency based on a difference between the resonance frequencies corresponding to the respective two gas cells.

Supplementary Note 3

(43) The atomic oscillator according to Supplementary Note 2, wherein the controller calculates, based on the difference between the resonance frequencies corresponding to the respective two gas cells, fluctuation amounts of the resonance frequencies with respect to temperature change of the alkali metal atoms encapsulated in the gas cells, and controls the oscillation frequency based on the calculated fluctuation amounts.

Supplementary Note 4

(44) The atomic oscillator according to Supplementary Note 2 or 3, wherein the controller controls the oscillation frequency based on the resonance frequency corresponding to one of the gas cells and a preset first parameter, and also changes the first parameter based on the difference between the resonance frequencies corresponding to the respective two gas cells to control the oscillation frequency based on the changed first parameter and the resonance frequency corresponding to the one gas cell.

Supplementary Note 5

(45) The atomic oscillator according to Supplementary Note 4, wherein the controller controls so as to generate a reference signal to be a reference for the irradiation light based on the oscillation frequency and a preset second parameter, and also changes the second parameter based on the difference between the resonance frequencies corresponding to the respective two gas cells to control so as to generate the reference signal based on the changed second parameter and the oscillation frequency.

Supplementary Note 6

(46) The atomic oscillator according to any of Supplementary Notes 1 to 5, wherein the two gas cells are configured to have different fluctuation amounts of the resonance frequencies with respect to temperature change.

Supplementary Note 7

(47) The atomic oscillator according to any of Supplementary Notes 1 to 6, wherein the two gas cells are configured with buffer gases of different types or pressures encapsulated in, respectively.

Supplementary Note 8

(48) A control method by an atomic oscillator, the atomic oscillator comprising: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a control apparatus that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies, the two gas cells being configured to have different characteristics of the corresponding resonance frequencies with respect to temperature, wherein the control method comprises, by the control apparatus, controlling the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

Supplementary Note 9

(49) 9. A control apparatus of an atomic oscillator, the atomic oscillator comprising: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and the control apparatus that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies, the two gas cells being configured to have different characteristics of the corresponding resonance frequencies with respect to temperature, wherein the control apparatus controls the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

Supplementary Note 10

(50) A computer program of an atomic oscillator, the atomic oscillator comprising: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a control apparatus that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an

oscillation frequency based on the determined resonance frequencies, the two gas cells being configured to have different characteristics of the corresponding resonance frequencies with respect to temperature, wherein the computer program comprises instructions for causing the control apparatus to execute processes to control the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells.

DESCRIPTION OF REFERENCE NUMERALS

(51) **1** light generator **11** laser **21** frequency measuring gas cell **22** temperature measuring gas cell **3** light detector **4** controller **41** resonance frequency and temperature shift amount calculator **42** current modulator controller **43** laser controller **44** frequency converter controller

Claims

1. An atomic oscillator comprising: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a controller that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies, wherein: the two gas cells are configured to have different characteristics of the corresponding resonance frequencies with respect to temperature, the controller controls the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells, the controller controls the oscillation frequency based on a difference between the resonance frequencies corresponding to the respective two gas cells, and the controller controls the oscillation frequency based on the resonance frequency corresponding to one of the gas cells and a preset first parameter, and also changes the first parameter based on the difference between the resonance frequencies corresponding to the respective two gas cells to control the oscillation frequency based on the changed first parameter and the resonance frequency corresponding to the one gas cell.
2. The atomic oscillator according to claim 1, wherein the controller calculates, based on the difference between the resonance frequencies corresponding to the respective two gas cells, fluctuation amounts of the resonance frequencies with respect to temperature change of the alkali metal atoms encapsulated in the gas cells, and controls the oscillation frequency based on the calculated fluctuation amounts.
3. The atomic oscillator according to claim 1, wherein the controller controls so as to generate a reference signal to be a reference for the irradiation light based on the oscillation frequency and a preset second parameter, and also changes the second parameter based on the difference between the resonance frequencies corresponding to the respective two gas cells to control so as to generate the reference signal based on the changed second parameter and the oscillation frequency.
4. The atomic oscillator according to claim 1, wherein the two gas cells are configured to have different fluctuation amounts of the resonance frequencies with respect to temperature change.
5. The atomic oscillator according to claim 1, wherein the two gas cells are configured with encapsulated buffer gases of different types or pressures.
6. A control method by an atomic oscillator, the atomic oscillator comprising: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a control apparatus that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies, the two gas cells being configured to have different characteristics of the corresponding resonance frequencies with respect to

temperature, wherein the control method comprises, by the control apparatus, controlling the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells, controlling the oscillation frequency based on a difference between the resonance frequencies corresponding to the respective two gas cells, and controlling the oscillation frequency based on the resonance frequency corresponding to one of the gas cells and a preset first parameter, and also changing the first parameter based on the difference between the resonance frequencies corresponding to the respective two gas cells to control the oscillation frequency based on the changed first parameter and the resonance frequency corresponding to the one gas cell.

7. A control apparatus of an atomic oscillator, the atomic oscillator comprising: two gas cells with alkali metal atoms encapsulated in, respectively; a light generator that irradiates each of the two gas cells with an irradiation light having at least two different frequency components; a light detector that detects transmitted lights transmitted by the two gas cells, respectively; and a control apparatus that determines resonance frequencies corresponding to the respective two gas cells based on light amounts of the transmitted lights by the respective two gas cells, and controls an oscillation frequency based on the determined resonance frequencies, the two gas cells being configured to have different characteristics of the corresponding resonance frequencies with respect to temperature, wherein the control apparatus controls the oscillation frequency based on a result of comparison between the resonance frequencies corresponding to the respective two gas cells, the control apparatus controls the oscillation frequency based on a difference between the resonance frequencies corresponding to the respective two gas cells, and the control apparatus controls the oscillation frequency based on the resonance frequency corresponding to one of the gas cells and a preset first parameter, and also changes the first parameter based on the difference between the resonance frequencies corresponding to the respective two gas cells to control the oscillation frequency based on the changed first parameter and the resonance frequency corresponding to the one gas cell.
